

Lecture 4: Compliment, Odds in Favour/Against, De Morgan's Law

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Topics of This Lecture: ✓

1. Properties ✓
2. Complement of Events ✓
3. De Morgans Law ✓
4. Subset ✓
5. Axioms of Probability ✓
6. Odds in Favour ✓
Odds in Against ✓
7. Problems ✓

1) Union and Intersection are :

1. Commutative: ✓

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

2. Associative: ✓

- $A \cup (B \cap C) = (A \cup B) \cap C$
- $A \cap (B \cup C) = (A \cap B) \cup C$

3. Distributive: ✓

- $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
- $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

2) Compliments of Events ($\bar{E}$):

Complement of events consists of all elements that are present in sample space (S) but not present within event (E).

$\bar{E}$ occurs if E doesn't occur

$$\bar{E} = S - E$$

$$(|\bar{E}|/|S|) = (|S|/|S|) - (|E|/|S|)$$

$$P(\bar{E}) = 1 - P(E)$$

$$E \subseteq S$$



$$\bar{E} = 1 - E$$

$$\bar{E} = S - E$$

$$P(\bar{H}) = ?$$

$$P(S) = 1$$

$$P(\bar{H}) = S - H = 1 - H$$

$$P(\bar{H}) = P(S) - P(H) = 1 - \frac{1}{2} = \frac{1}{2}$$



$$S - H = \bar{H}$$
$$S - T = \bar{T}$$

3) De Morgan's Law:

- $\overline{A \cup B} = \bar{A} \cap \bar{B}$
- $\overline{A \cap B} = \bar{A} \cup \bar{B}$

4) Subset:

Let we have two events E & F.

If all outcomes in E are also present in F
Then E is contained in F

We can say E is subset of F

$$E \subseteq F$$

5) Axioms of Probability:

- **Axiom 1:**
 $0 \leq P(E) \leq 1$

- **Axiom 2:**
 $P(S) = 1$

- **Axiom 3:**
For any sequence of mutually exclusive events
 $E_1, E_2, E_3, \dots, E_n$ ($E_i \cap E_j = \emptyset$ for all i, j)

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i)$$

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$$

Q. A total of 40% Indian use Gpay. ✓ ✓

10 % use PhonePay ✓ ✓

5 % use both ✓

What % of Indian neither use Gpay nor PhonePay?

✓ Union

✓ Intersection

✓ Complement of Event

$$P(G) = 0.4$$

$$P(P) = 0.1$$

$$P(G \cap P) = 0.05 \checkmark$$

X E

X P

$$\begin{aligned} P(G \cup P) &= |G| + |P| - |G \cap P| \\ &= 0.4 + 0.1 - 0.05 \\ &= 0.5 - 0.05 = 0.45 \end{aligned}$$

$$P(\overline{G \cup P}) = ?$$

$$\boxed{\overline{E} = S - E}$$

$$\begin{aligned} P(\overline{G \cup P}) &= 1 - 0.45 \\ &= 0.55 \end{aligned}$$

55% population
X Gpay
X Ppay.

6) Odds in Favour:

Ratio of no. of favourable outcomes to no. of unfavourable outcomes of a particular event is called odds in favour of an event.

$$\text{Odds in favour of event } A = P(A)/P(\bar{A}) = P(A)/(1-P(A)) = |A|/|\bar{A}|$$

Odds in Against:

Ratio of no. of unfavourable outcomes to no. of favourable outcomes of a particular event is called odds in against of an event.

$$\text{Odds in against of event } A = P(\bar{A})/P(A) = 1-P(A)/P(A) = |\bar{A}|/|A|$$

Q2. Find the odds in favour & odds in against of throwing a die and getting a 6?

$E = \text{getting a 6}$

$$|E| = 1$$

$$\bar{E} = |S| - |E|$$

$$|\bar{E}| = 5$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(E) = \frac{|E|}{|S|} = \frac{1}{6}$$

$$|S| = 6$$

$$|E| = 1$$

$$P(\bar{E}) = \frac{|\bar{E}|}{|S|} = \frac{5}{6}$$

$$|\bar{E}| = 6 - 1 = 5$$

$$\text{odds in favour of } E = \frac{|E|}{|\bar{E}|} = \frac{1}{5}$$

$$\text{odds in against of } E = \frac{|\bar{E}|}{|E|} = \frac{5}{1}$$

Q3. Odds in favour of India winning the world cup is 4:3 ✓

And odds in against of New Zealand winning is 2:6 ✓

Find the probability of :

- India winning the cup ✓
- New Zealand winning the cup ✓

$\frac{I}{N}$

$$\text{odds in favour of } I = \frac{|I|}{|\bar{I}|} = \frac{4}{3}$$

$$\text{odds in against of } N = \frac{|\bar{N}|}{|N|} = \frac{2}{6}$$

$$\text{odds in favour of } I = \frac{|I|}{|\bar{I}|} = \frac{P(I)}{1 - P(I)} = \frac{4}{3}$$

$$\Rightarrow 3P(I) = 4 - 4P(I)$$

$$\Rightarrow 2 P(I) = 4 - 4P(J)$$

$$\Rightarrow 7 P(I) = 4$$

$$\Rightarrow P(I) = \frac{4}{7}$$

$$\text{Odds in favour of } N = \frac{|N|}{|I|} = \frac{1 - P(I)}{P(I)} = \left(\frac{2}{6} \right)$$

$$\Rightarrow 6 - 6P(N) = 2P(N)$$

$$\Rightarrow 4P(N) = 6$$

$$\Rightarrow P(N) = \frac{6}{4} = \left(\frac{3}{2} \right)$$